CHAPTER: 2

The Economic Problems

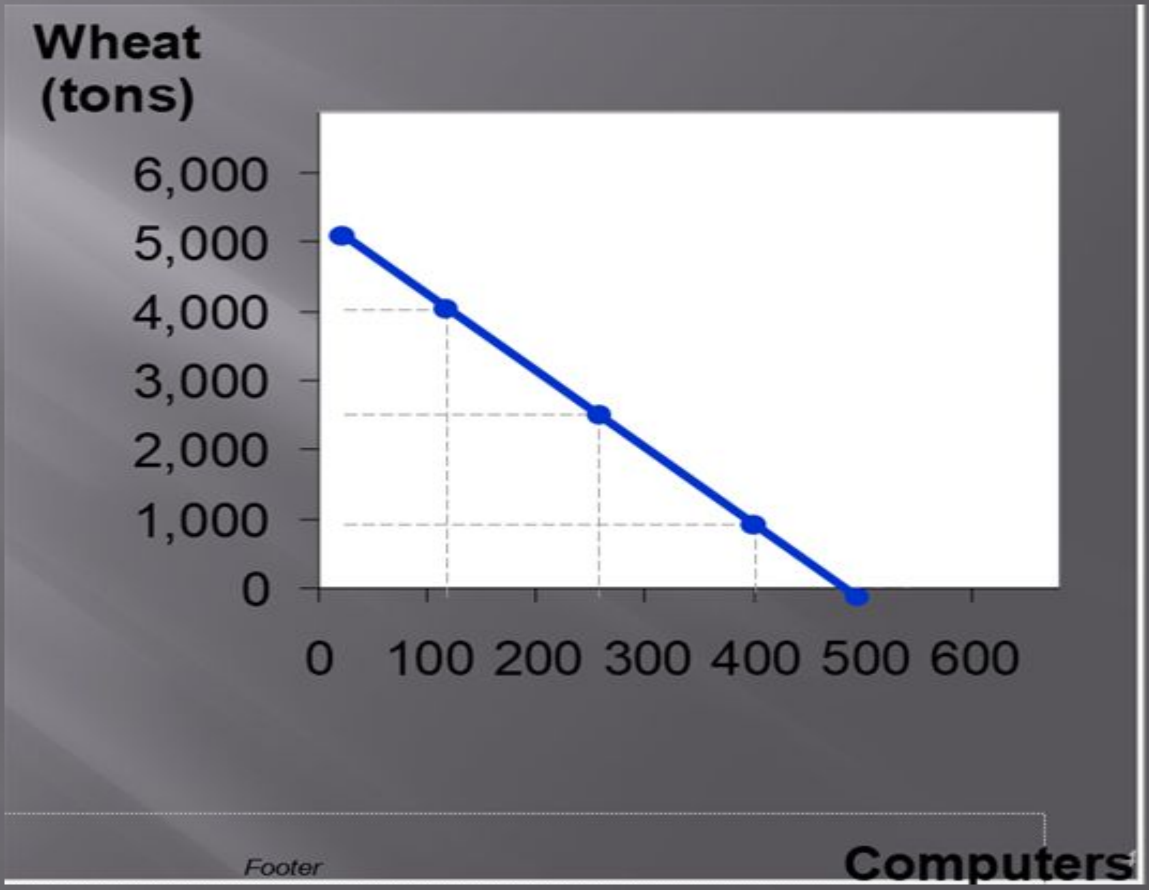
The PPF

- Production possibilities frontier
  - A graph: combinations of output that the economy can possibly produce
  - Given the available
    - Factors of production and technology
  - Example:
    - Two goods: computers and wheat
    - One resource: labor (measured in hours)
    - Economy has 50,000 labor hours per month available for production

Producing one computer requires 100 hours of labor
Producing one ton of wheat requires 10 hours labor.

	Employment of labor hours		Production	
	Computers	Wheat	Computers	Wheat
A	50,000	0	500	0
B	40,000	10,000	400	1,000
C	25,000	25,000	250	2,500
D	10,000	40,000	100	4,000
E	0	50,000	0	5,000

Point on graph	Production	
	Com-puters	Wheat
A	500	0
B	400	1,000
C	250	2,500
D	100	4,000
E	0	5,000

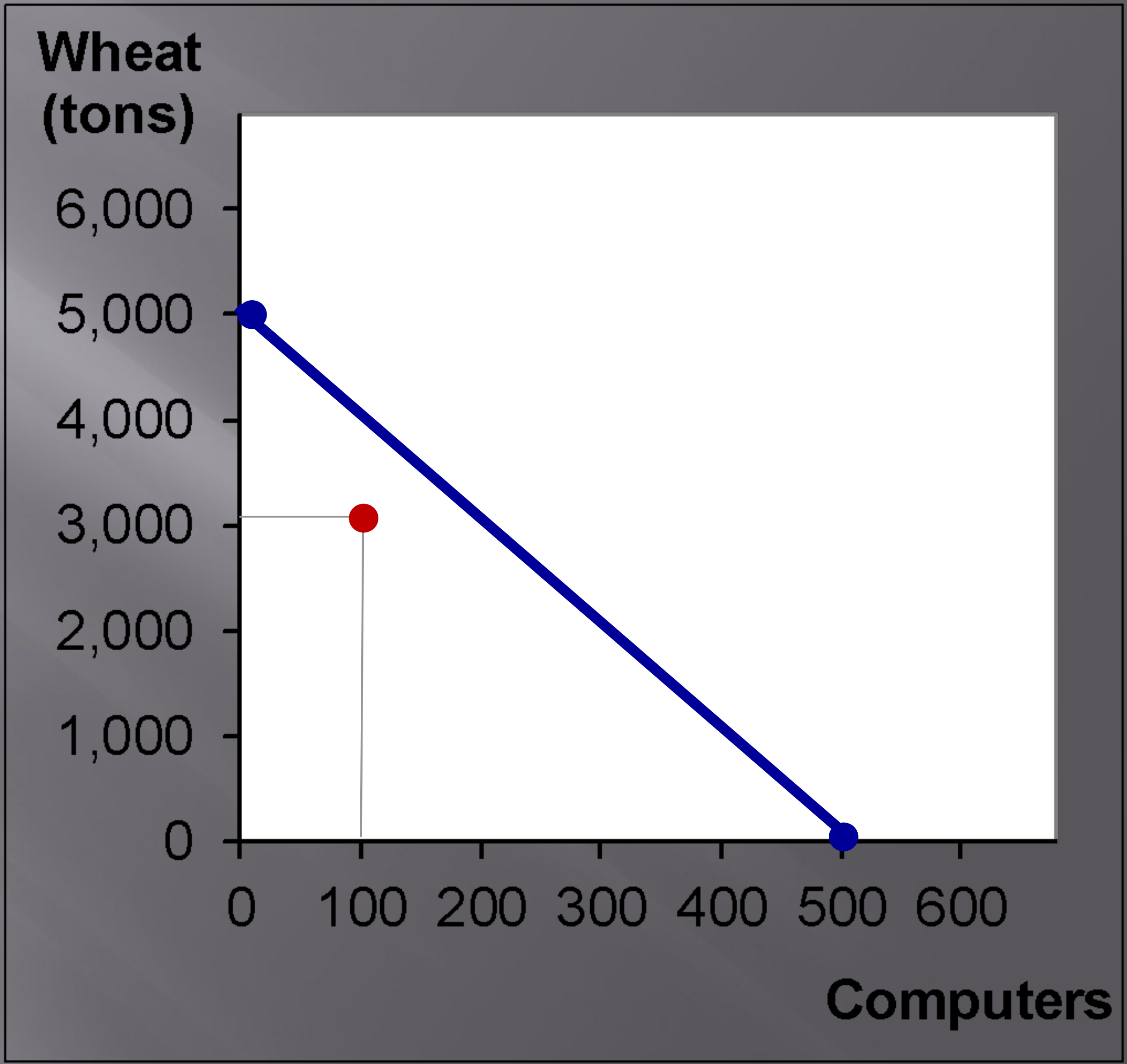


Production Possibilities Frontier

- PPF illustrates the concept of:
- i)Production efficiency
- ii)Trade-off
- iii) Opportunity cost
- iv)Economic growth

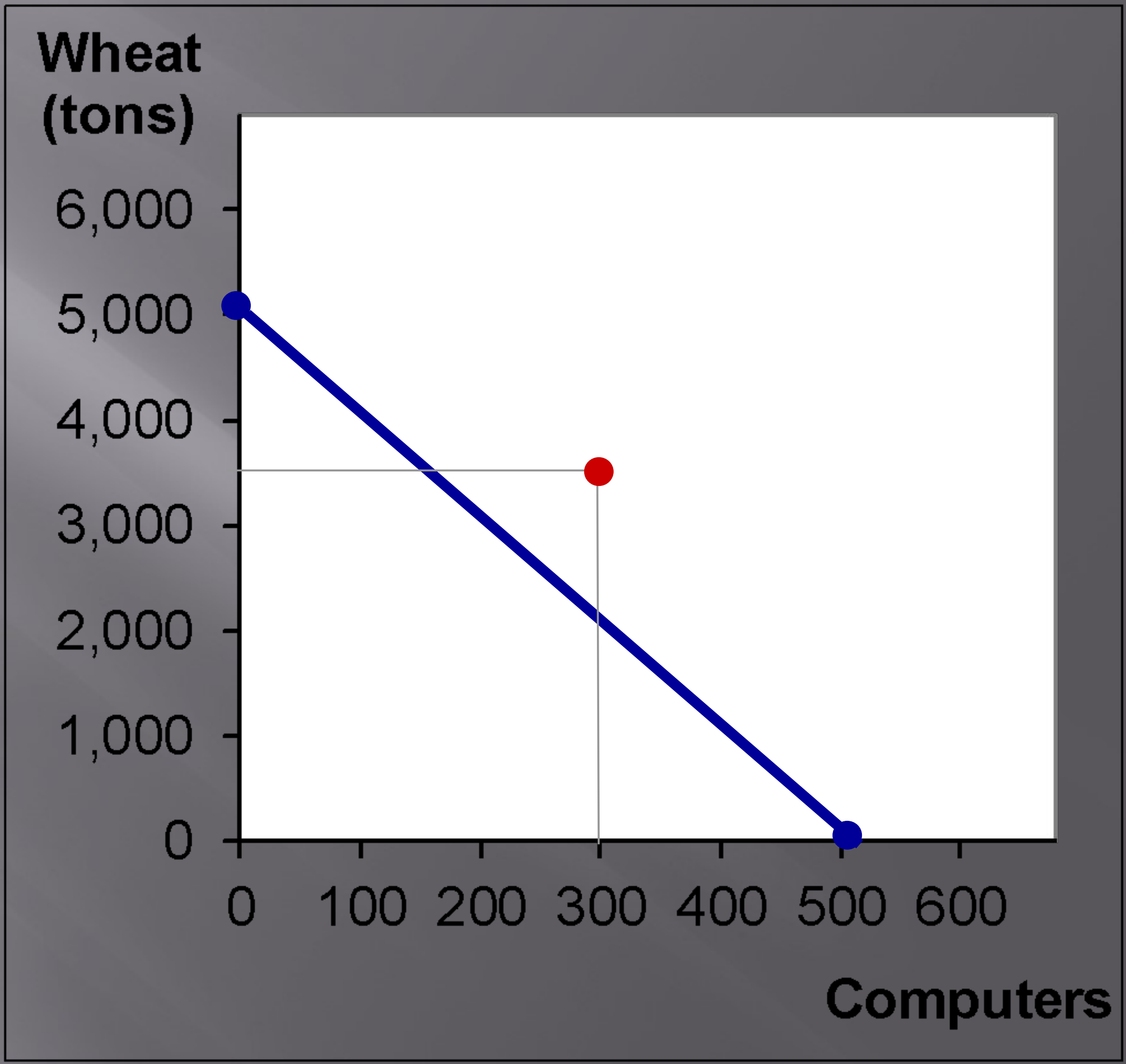
Points off the PPF

- Point F: 100 computers, 3000 tons wheat
- Requires 40,000 hours of labor
- Possible but not efficient: could get more of either good without sacrificing any of the other



Points off the PPF

- Point G: 300 computers, 3500 tons wheat
- Requires 65,000 hours of labor.
- Not possible because the economy only has 50,000 hours



The PPF & Efficiency

- Points on the PPF (like A – E): possible
  - Efficient: all resources are fully utilized
- Points under the PPF (like F): possible
  - Not efficient: some resources are underutilized (e.g., workers unemployed, factories idle)
- Points above the PPF (like G)
  - Not possible

Tradeoff Along the PPF

- **Tradeoff Along the PPF**
  - A choice *along* the PPF involves a *tradeoff*.
  - At any given time, we have a fixed amount of labor, land, capital and entrepreneurship and a given state of technology.
  - We can employ these resources to produce limited goods and services. That is if we produce more wheat, we need to reduce the production of cola and vice versa.

Opportunity Cost

The trade-offs that we make involve a cost. In economics it is called Opportunity cost.

- The **opportunity cost** of an action is the highest- valued alternative forgone.
- We can calculate the opportunity cost from the PPF-
- The opportunity cost of producing an additional computer is the wheat we must forgo. Similarly, the opportunity cost of producing an additional ton of wheat is the quantity of computer we must forgo.

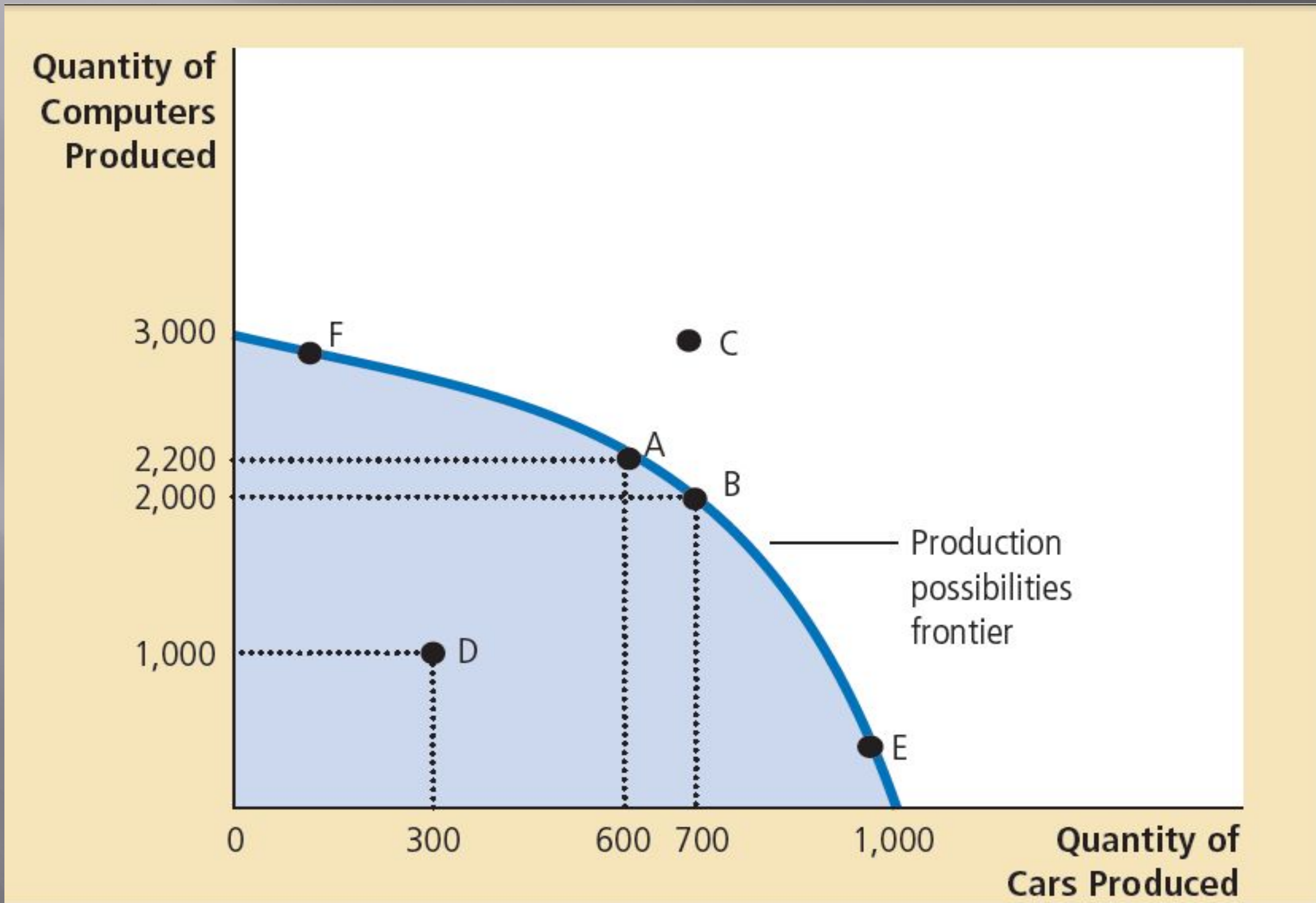
Opportunity Cost is a Ratio

- It is the decrease in the quantity produced of one good divided by the increase in the quantity produced of another good as we move along the production possibilities frontier.

The Shape of the PPF

- The PPF could be a straight line(example 1), or bow-shaped(example 2)
- Depends on what happens to opportunity cost as economy shifts resources from one industry to the other.
  - If opp. cost of a good rises as the economy produces more of the good, PPF is bow-shaped.
  - If opp. cost remains constant, PPF is a straight line.

Figure: The Production Possibilities Frontier



The Production Possibilities Frontier cont..

- PPF shows the opportunity cost of 1good in terms of the other.
- This is measured by the slope of PPF.
- In the example, The slope varies across different points.(Not constant)
- Opportunity cost is higher at point E, where the economy produces more cars than computers.(Frontier is steep)
- In contrast, it is lower at point F, where economy produces more computers than cars(Frontier is relatively flatter)

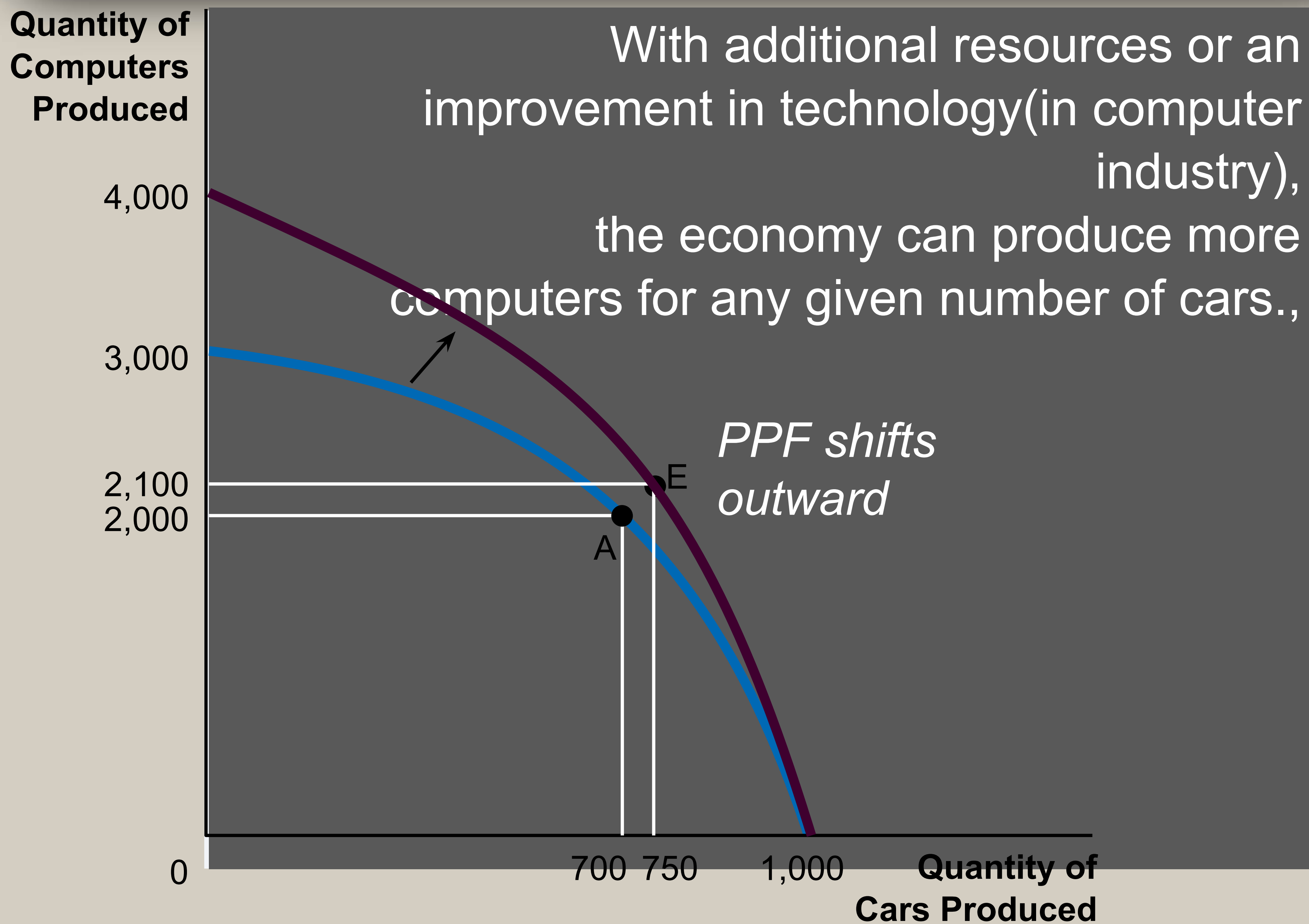
Economic Growth & PPF

- Economic growth occurs for **two** reasons:
- **i)Technological change**: refers to new and better way of producing goods and services(I;e Same product is produced using advanced technology)
- **ii) Capital accumulation**: Increase in the amount of capital resources(New factory, buildings,). Also includes human capital.

Economic Growth &PPF

- Thus an increase in capital resources or advancement of technology-
- Causes economic growth.
- The country can produce more products and services then before.
- The PPF shifts rightward

Figure 3 A Shift in the Production Possibilities Frontier



Increase in Resources(In Both Sectors)